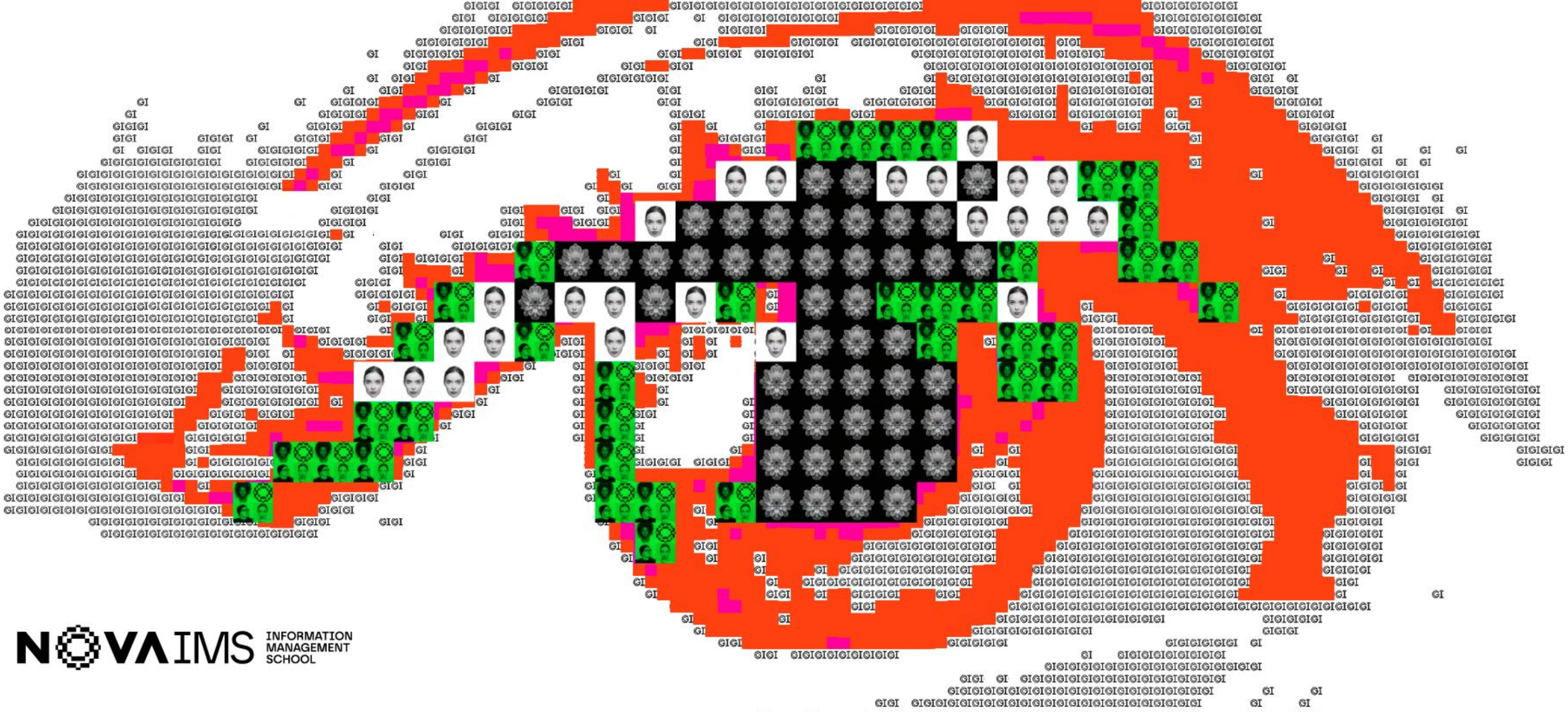


NOVA IMS INFORMATION MANAGEMENT SCHOOL

Big Data Analytics with Spark

Scalable Sentiment Intelligence with Apache Spark

Big Data Analytics | Spring Semester 2025/26 | Group 3

01

Business Challenge: Why Spark?

Why social media data is difficult to monitor at scale and how Spark solves that problem

02

Data & Processing Approach

How comments and research data were prepared for analysis

03

Insights & Solutions

Key findings from sentiment and research analytics; prototype tools developed with Spark

04

Business Impact and Next Steps

How the client can turn the prototype into value

Why does this problem require Spark?

VOLUME

Millions of comments generated daily across thousands of videos. Traditional tools hit memory and time limits.

VELOCITY

New comments are generated continuously and must be analysed quickly.

VARIETY

Multi-language text with emoji, slang, and mixed sentiment. Preprocessing must handle all formats at scale.

VERACITY

Social media data contains noise, spam and inconsistent information that must be filtered before analysis.

Data & Processing Approach

Where did the data come from and how was it prepared?

YouTube Comments Dataset

We analysed 1M+ labelled YouTube comments sourced from Hugging Face.

Each comment was classified as Positive, Negative or Neutral.

The dataset supported sentiment insights, model training and real-time monitoring.

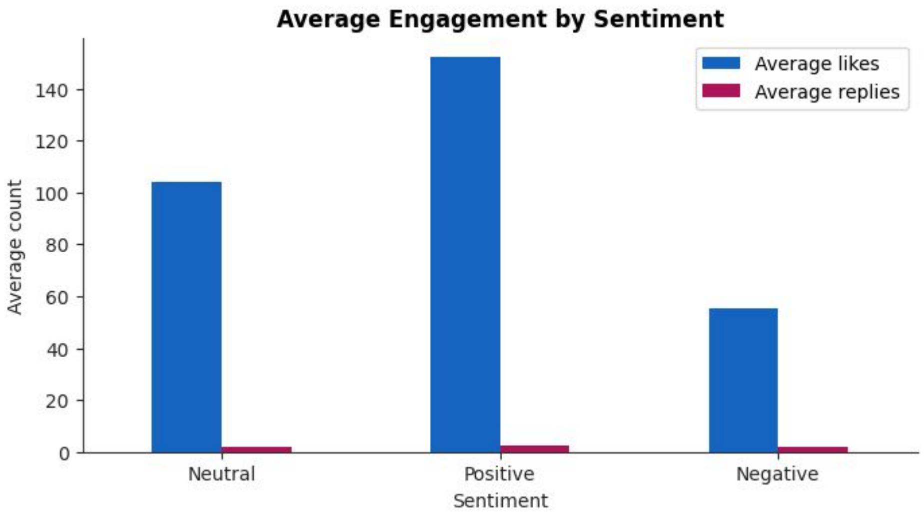
Research Network Dataset

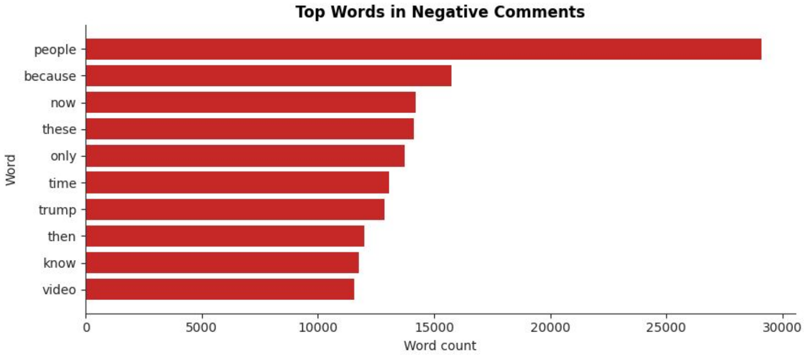
We analysed a large research network containing papers, authors, venues and citation relationships.

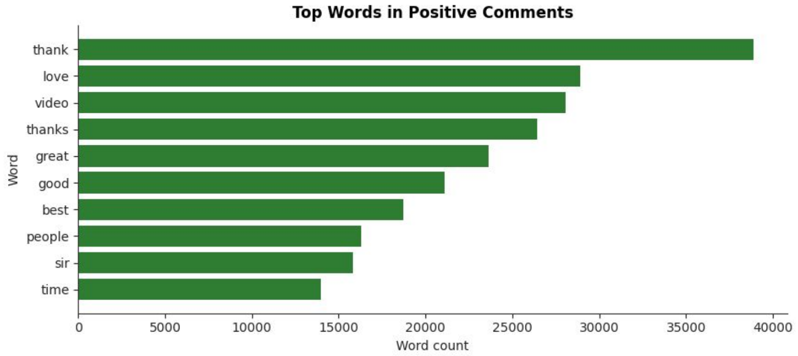
The dataset was used to map the Sentiment Analysis research ecosystem and identify communities, cross-domain researchers and knowledge connections.

Insights and Solutions

Initial Findings







Spark Structured Streaming

01 › **02** › **03** › **04**

- **01** — YouTube API
- **02** — Spark Streaming
- **03** — SVM Model
- **04** — MongoDB Database Storage

@sal_s_sal	Remove cube	Neutral
@andreapessoa7124	UHUUUUUUUUUUU!!! TOP!!! 🤩	Positive
@Cristãodochat	A gameplay parece boa, gráfico bom e direção interessante, mas... não é mais...	Positive

- Identify viral negative threads before they escalate to issues
- Track how community sentiment evolves across a campaign lifecycle

Business Value - Streaming

- **Proactive reputation management**: Detecting viral negative threads before they escalate allows organisations to intervene early.

- **Spark is fundamental at this scale**: Processing continuous comment streams with ML inference requires distributed, in-memory computation; no single-node solution could meet this latency and throughput requirement without Spark.

- **Campaign optimization in real time**: Tracking sentiment evolution enables marketers to adjust messaging or pause underperforming content while it still matters

Research Network Intelligence

17k+
Research Communities Identified

≈ 5k
Cross-Domain Researchers

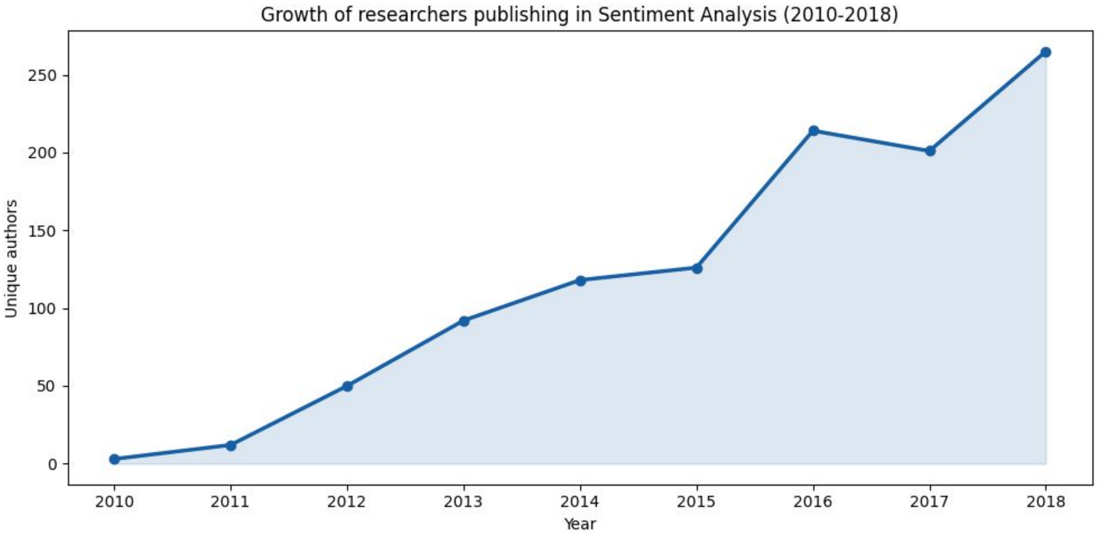
500k+
Knowledge Connections

Understanding where expertise and innovation are concentrated helps organisations identify trusted knowledge sources, emerging trends and promising areas for future investment.

Evolution of Sentiment Analysis Research

<u>Key insight</u>
Sentiment Analysis has evolved from a niche topic into a rapidly growing research field.

<u>Evidence</u>
Active researchers increased from 3 to 265 between 2010 and 2018.



<u>A larger research community increases the availability of proven methods, specialised expertise and business-ready applications.</u>

What now?

Identify emerging trends and negative sentiment before they become larger issues.

Spark provides a foundation that can scale from prototype datasets to production environments.

Expand data sources, deploy dashboards and alerts, and integrate real-time monitoring into business workflows.

Spark transforms large-scale social media data into actionable insights, enabling faster and more informed decisions.